NAME: _____ NetID:_____

INFO 4220 – Networks II: Market Design
Spring 2024 – Prelim 1

Instructions:

- **Put your name and NetID in every page (do it now). Exams will be scanned and processed using gradescope. If you don't put your name and NetID in the designated spaces, we may never be able to link your answers to you.**
- **Please do not separate the pages. If you do, staple them back at the end.**
- **Time limit:** You have **60 minutes** to complete this exam.
- **Asking questions:** Read all the questions now and make any questions you may have at the beginning to avoid disturbing others. If you think something is wrong or some information is missing, make the assumptions that you feel are necessary to answer and state them along with your answer.
- **Cheat Sheet:** Only this printed exam and a 1-page cheat sheet are allowed in your table during the exam. Only 1 side of the cheat sheet can be written (handwritten, not printed), your name and netid must be written on top of the cheat sheet, your cheat sheet must be your own creation.
- **Devices:** You cannot use any device (cell phone, smart watch, calculator, computer, etc.) during the exam.
- **Academic Integrity:** Any academic integrity violation will result in an academic integrity hearing in which a penalty consistent with Cornell's code of academic integrity will be determined.
- **Write clearly and show all your steps.** If you need more space for a question, use the last page.

Problem 1 (18 points, 3 points for each question)

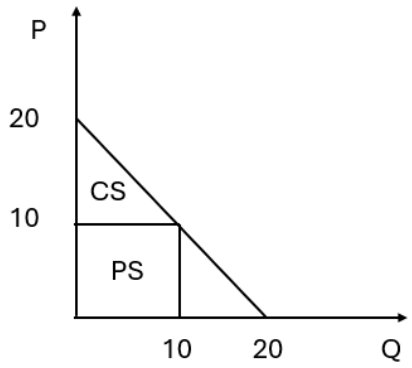
1.1. In a market for widgets, the demand curve is given by $Q_D = 20 - P$, and the supply curve by $Q_S = 10$. Find the equilibrium price, the consumer surplus, and the producer surplus in the market equilibrium. Make a graph showing the demand curve, supply curve, market equilibrium, consumer surplus and producer surplus (the graph does not need to be to scale).

Solution:

Equilibrium: Q* = 10 (0.5pt), P* = 10 (0.5pt)

CS = $\frac{10*10}{2} = 50$ (0.5pt), PS = $10 * 10 = 100$ (0.5pt) (It's not required to show the calcuation, showing the results for CS and PS is enough here.)

(Graph: 1pt)



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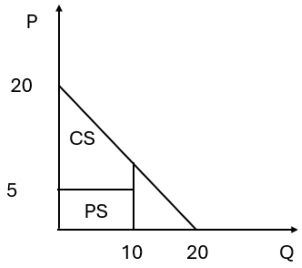
1.2 What if in the market from problem 1.1 the government imposes a price cap of $5. Find the new quantity to be traded, the producer surplus, consumer surplus, and deadweight loss produced by the price cap. Show these quantities in a new graph.

Solution:

Equilibrium: Q* = 10 (0.5pt), P* = 5 (0.5pt)

CS = 100 , PS = $5 * 10 = 50$ (1pt)

DWL= 0 (1pt)



1.3 In an assignment problem you need to assign 5 different types of food to 5 animals. None of the animals own any food at the beginning of the problem (i.e. there are no endowments). You decide to use a serial dictatorship to assign the foods. Your friend, who is also enrolled in this class, says that after you run the serial dictatorship you can use the top trading cycle (TTC) algorithm to improve on the assignments in the following way: You run the TTC algorithm considering that the endowment of each animal is the food they were assigned in the serial dictatorship. Prove that your friend is wrong.

Solution:

Running a TTC after serial dictatorship will not help, because no one would like to trade after the allocation from TTC.

In every step in the serial dictatorship, each animal chose the fruit they preferred among those available. Thus, there will no trades possible in TTC, because:

- Everyone that chose before has a fruit that they like better
- They will not want to trade with anyone that chooses later

1pt: take the most preferable object among the remaining ones in serial dictatorship, 2pts: clear explanation for why there will be no trade for TTC after serial dictatorship

Alternative solution (3pts): Serial dictatorship always produces efficient assignments, so no one would like to trade/change assignments

1.4 In a medical match problem there are 3 hospitals (h_1, h_2, h_3) and 4 doctors $(d_1, d_2, d_3. d_4)$. Hospitals h_1 and h_2 have a capacity of 2, while h_3 has a capacity of 1. The preferences of hospitals and doctors are in the table below:

h_1	h_2	h_3
d_1	d_1	d_1
d_2	d_2	d_2
d_3	d_3	d_3
d_4	d_4	d_4

d_1	d_2	d_3	d_4
h_3	h_3	h_3	h_3
h_1	h_2	h_1	h_2
h_2	h_1	h_2	h_1

Use the deferred acceptance algorithm with hospitals proposing to find a matching in this problem:

Solution:

Step 1: h1 -> ~~d1~~, ~~d2~~; h2 -> ~~d1~~, d2; h3 -> d1.

d1 rejects h1, h2, d2 rejects h1. d1 is temporarily matched to h3, d2 is temporarily matched to h2.

Step 2: h1 -> d3, d4; h2 -> ~~d3~~

d3 rejects h2. d3 is temporarily matched to h1, d4 is temporarily matched to h1.

Step 3: h2 -> d4

d4 prefers h2 over h1, so rejects h1 and match with h2. END

$\mu(h_1) = \{d_3\}, \mu(h_2) = \{d_2, d_4\}, \mu(h_3) = \{d_1\}$

1pt for each step x 3.

Only 1pt for the whole question if only provides the final matching result without showing the process.

1.5 Is there any way any hospital in problem 1.4 could misrepresent their preference to obtain a matching they prefer to what you found in 1.4?

Solution:

No. According to the rural hospital theorem, if a hospital does not fill its vacancies at some stable matching, it will always be matched to the same set of doctors at all stable matchings. So, for h_1, they will always be matched to d_3 and no one else. And note h_3 already has their top choice (doesn't want to change their preference list), and whatever hospital 2 does, they will never be able to get d_1 away from h_3 (because it is d_1 preferred hospital)

0 points if says D.A. with hospitals proposing is strategy proof, because it is not. 0 points if says No without any explanation.

2pt for answering "No, there is no better way" and providing a reasoned answer based on preferences

0.5pt for mentioned rural hospital theorem.

0.5 pt for explanation of the theorem and elaborate it with results from 1.4 (h1 will always have vacancies or other possible explanation).

1.6 Ash, May, Misty, and Nemona are trying to assign Pokemons for their next adventure. We will treat Pokemons as objects (i.e. they don't have preferences), and consider the following initial endowment: Ash – Charmander, May – Pikachu, Misty – Squirtle, Nemona – Bulbasaur. If the preferences for Pokemons are as in the table below, explain and show why the top trading cycle algorithm is a better choice than a serial dictatorship in this problem:

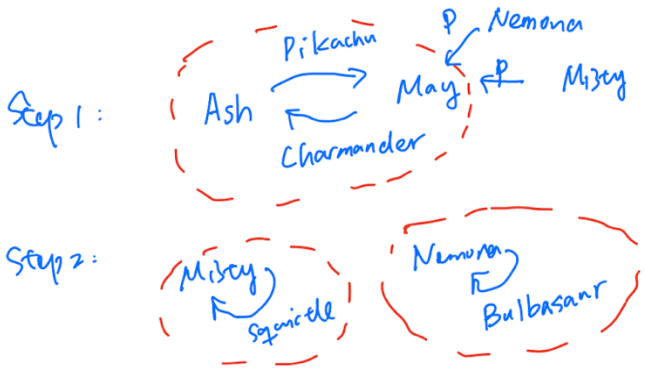
Ash	**May**	**Misty**	**Nemona**
Pikachu	Charmander	Pikachu	Pikachu
Charmander	Squirtle	Squirtle	Charmander
Squirtle	Bulbasaur	Bulbasaur	Bulbasaur
Bulbasaur	Pikachu	Charmander	Squirtle

Solution:

If using serial dictatorship, it will produce results that are not individually rational.

If we run serial dictatorship with the order of (Nemona, Misty, May, Ash), the result will be: Nemona – Pikachu, Misty – Squirtle, May – Charmander, Ash – Bulbasaur. Ash will end up with Bulbasaur which is worse than their initial endowment. **(1pt for showing a case from serial dictatorship could generate non-individually-rational results, any other order that could demonstrate the property also counts.)**

If using TTC with **private endowment**, the result is as below.

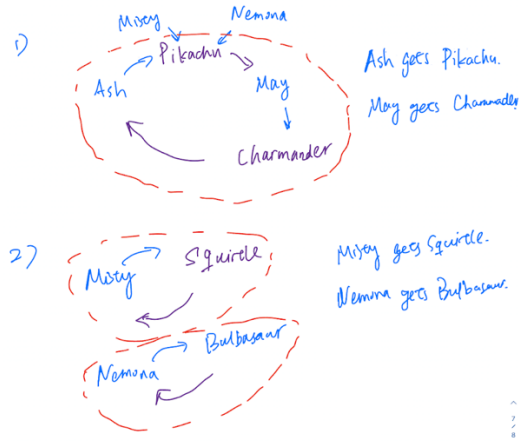


After TTC, Ash get Pikachu, May get Charmander, Mistry get Squirtle, and Nemona get Bulbasaur.

(1pt for TTC process and result)

TTC is better because it produces **individually rational** and efficient solution while serial dictatorship may not. **(1pt)**

Note: If running TTC with public endowment as below (individuals pointing to objects), the algorithm result will be the same for this question. However, since we treat Pokemons as objects (i.e. they don't have preferences), and consider the initial endowment (private endowment), you should use TTC with private endowment with individual points to the individual in the graph. We do not deduct points for the different graph representation this time, but please make sure you have the right presentation in the future to get full credits.



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Problem 2 (10 points, 5 points each letter)

Adam and Beatrice work in the same company. They are assigned to draft a report about a project that they have been working on for the last couple of months. They will work separately on two different sections of this report, and they will not be able to communicate with each other during this time. While Beatrice knows that Adam is a hard worker, Adam does not know whether she will be **Not Focused**, namely putting the minimal effort on this, or **not focused**, namely putting a high effort on the report. Adam will receive a signal about whether Beatrice is Not Focused or focused. This signal is the probability that Beatrice is Not Focused, and we will denote it with p.

Adam has to decide whether to draft the entire report ('Complete') or to draft only the section that is assigned to him. When he chooses 'Complete', he faces an extra cost $c > 0$ for having worked more than expected. The matrixes below show the payoff that both receive, one for each Beatrice's type.

Beatrice is **Not Focused**			
		Beatrice	
		Draft	Free Ride
Adam	Complete	$8 - c$, 5	$6 - c$, 6
	Partial	6 , 6	5 , 5

Beatrice is **Focused**			
		Beatrice	
		Draft	Free Ride
Adam	Complete	$10 - c$, 8	$6 - c$, 6
	Partial	10 , 10	5 , 5

a) What are the conditions on the cost c that will make Adam decide to write the draft himself in each of the possible states of Beatrice (i.e., he chooses 'Complete')? (5 points)

Solution:

In order for Adam to choose action 'Complete', it must be that the utility that Adam receives from choosing the action is higher than the one he will obtain by choosing Partial.

We first consider the case in which Beatrice is Not Focused **and** fix Beatrice's action. Then, Adam will compare the following utilities:

$u_{Adam}(Complete, Draft|\text{Not Focused}) = 8 - c \; vs. \; 6 = u_{Adam}(Partial, Draft|\text{Not Focused})$

In this case, Adam will choose 'Complete' if $c < 2$. **(1pt)**

$u_{Adam}(Complete, FreeRide|\text{Not Focused}) = 6 - c \; vs. \; 5 = u_{Adam}(Partial, FreeRide|\text{Not Focused})$ In this case, Adam will choose 'Complete' if $c < 1$. **(1pt)**

Next, we consider the case in which Beatrice is focused **and** fix Beatrice's action. Adam will compare:

$u_A(Complete, Draft|Focused) = 10 - c \; vs. 10 = u_A(Partial, Draft|Focused)$

In this case, since we have assumed that $c > 0$, Adam will always get a higher utility from choosing 'Partial' when Beatrice is Focused, **and** we assume she will draft. **(1pt)**

$u_A(Complete, FreeRide|Focused) = 6 - c \; vs. 5 = \; u_A(Partial, FreeRide|Focused)$

In this case, Adam chooses 'Complete' if $c < 1$. **(1pt)**

We can conclude that if Beatrice is Not Focused, Adam will choose Complete if $c < 1$ (and $c > 0$). If Beatrice is focused, having $c < 1$ is not sufficient to make sure that Adam will strongly prefer to

complete rather than choosing 'partial' (there is no value of c (which is > 0) that makes Completing a dominant strategy for Alex when Beatrice is Not Focused. **(1pt)**

Rubric: 2 pts for analysis on if Beatrice is Not Focused and 2pts for analysis on if Beatrice is focused, get partial credits if the analysis is not sufficient; 1pts for the conclusion

b) Assume that $c = 1$, what are the conditions on the probability that Beatrice is Not Focused (p) such that the following strategy profile $(Partial; (Draft, Draft))$ is a Bayes Nash Equilibrium? (5 points)

Solution:

If $c = 1$, the matrix of payoffs becomes:

Beatrice is Not Focused			
		Beatrice	
		Draft	Free Ride
Adam	Complete	**7**,5	**5**, **6**
	Partial	6, **6**	**5**, 5

Beatrice is **Focused**			
		Beatrice	
		Draft	Free Ride
Adam	Complete	9, 8	**5**, 6
	Partial	**10**, **10**	**5**, 5

(1pt for payoff when c=1. If the steps below are corrected, students also get credits. It's not required to write out the matrix.)

We are asked to find $p = \Pr(Beatrice = Not\ Focused)$ such that $(Partial; (Draft, Draft))$ is a Bayes Nash Equilibrium. We focus on the expected payoff of Adam fixing Beatrice's actions to the strategy profile $(Draft, Draft)$. We must have that the expected payoff when he chooses 'Partial' is higher than the expected payoff when he chooses 'Complete'.

$E[u_A(Complete; (Draft, Draft))] = 7p + (1-p)9 = 9 - 2p$ **(1pt)**

$E[u_A(Partial; (Draft, Draft))] = 6p + (1-p)10 = 10 - 4p$ **(1pt)**

The first equation is higher than the second if $9 - 2p < 10 - 4p$ **(1pt),** namely when $p < 0.5$ **(1pt).** In other terms, if Adam believes that Beatrice is less likely (less than 50% in probability) to be Not Focused than Focused, that strategy profile is a Bayes Nash Equilibrium.

Note that if we assume Adam chooses partial, then Beatrice does not want to deviate from the strategy (Draft, Draft).

Rubric: 3pts for the expected payoff when c=1, 2pts for the analysis on finding BNE (1pt on the condition and 1pt on the result)

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Problem 3 (12 points, 3 points each letter)

Four researchers—Alice, Bob, Carol, and Denis—are to be allocated presentation slots at an economics conference. There are four slots in total: A and B in the morning, and C and D in the afternoon.

Alice	Bob	Carol	Denis
C	D	B	A
A	B	D	B
B	C	A	D
D	A	C	C

a) Use a Serial Dictatorship to find the assignment. Follow the order of dictators: Carol, Denis, Alice, Bob. Does the order matter for the assignment result in general (i.e. not only for this problem)? Does the order influence the efficiency and strategyproof-ness of the algorithm? Explain why.

Solution:

$u(Alice) = C, u(Bob) = D, u(Carol) = B, u(Denis) = A$ **(1pt)**

In this case, the order doesn't matter. However, in general, the order matters for the assignment results **(1pt for the general case)**. Those who can choose their option earlier would always get better options. And this may introduce the fairness issue. However, it does not influence the efficiency and strategyproofness of the algorithm. There is no other assignment that can make an individual better off while not make others worse off, and participants has no interests to lie. **(1pt for efficiency and strategyproof)**

Now, assume that because they were the first two researchers to submit their papers, Slot A was pre-assigned to Alice and Slot B was pre-assigned to Bob. Carol and Denis, having submitted their papers later, do not have a pre-assigned slot. Use this assumption for b), c) and d).

b) The organizers decide to assign the slots using a Random Serial Dictatorship with Squatting Rights. Bob decides not to participate and keep his pre-assigned slot. Should Alice participate in the assignment or just keep her pre-assigned slot? Why?

Solution

If Alice participates:

Bob decides not to participate. $u(Bob) = B, u(Carol) = D$ (since B is taken by Bob), $u(Denis) = A$, $u(Alice) = C$. **(1pt for results if Alice participates)**

Alice is better off than in the case in which she does not participate. Thus, Alice should participate in the new assignment. **(1pt for conclusion and 1pt for explanation)**

c) Assume that Alice decides to participate in the assignment. Is the assignment from b) efficient? Why or why not?

Solution: It's not efficient. **(2pt for conclusion)** Both Bob and Carol can be better off if they switch the slot with each other. **(1pt for explanation)**

d) To ensure an equitable distribution of presentation slots among all researchers, the committee seeks to adopt an algorithm that assures strategy-proofness, efficiency, and individual rationality. Choose an algorithm (of the ones we discussed in class) that fulfils these criteria and apply it to assign presentation slots to researchers. If a random order of individuals (or priority ordering) is required to initiate the process, use the order: Carol, Denis, Alice, Bob.

Solution:

For this problem, it is correct to use either YRMH-IGYT or TTC with mixed-endowment to solve the problem.

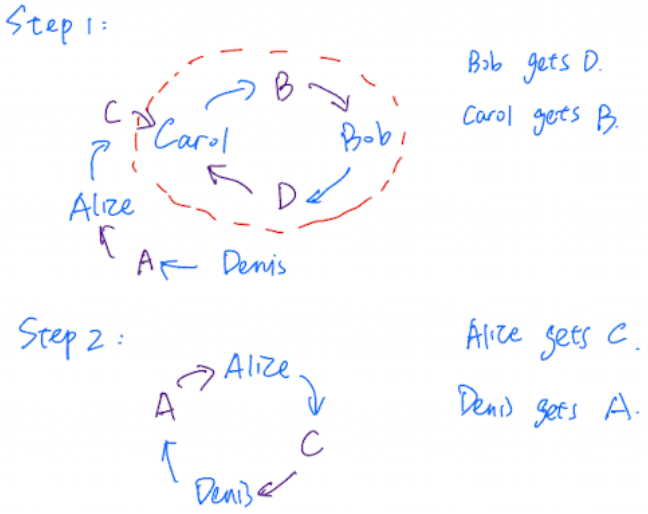
Option 1: YRMH-IGYT

Order of dictators: Carol, Denis, Alice, Bob

- Carol chooses B, but existing researcher Bob has not chosen a slot yet. Thus, the new order is (Bob, Carol, Denis, Alice). We resume from Bob.
- Bob chooses D, then Carol chooses B (note B is vacant after Bob has chosen D).
- Denis chooses A, but existing researcher Alice has not chosen a slot yet. Thus, the new order is (Bob, Carol, Alice, Denis). We resume from Alice.
- Alice chooses C, then Denis chooses A.
- So the final assignment is: $u(Alice) = C, u(Bob) = D, u(Carol) = B, u(Denis) = A$

2pts for the process of YRMH-IGYT, 1pt for the final assignment result.

Option 2: TTC with mixed-endowment



So the final assignment is: $u(Alice) = C, u(Bob) = D, u(Carol) = B, u(Denis) = A$

2pts for the process of TTC (1pt for each step), 1pt for the final assignment result.